

KELT-23Ab

R-2250

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KELT-23A_250620 · created 2026-08-15T23:56:03+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460844.47848 +/- 0.00098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.125 +/- 0.023
Transit depth (Rp/Rs)^2	1.57 +/- 0.57 [%]
Orbital Inclination (inc)	87.46 +/- 1.21
Airmass coefficient 1 (a1)	4.564 +/- 0.034
Airmass coefficient 2 (a2)	-1.24 +/- 0.4
Scatter in the residuals of the lightcurve fit is	0.75 %
Best Comparison Star	#1 - [78, 417]
Optimal Aperture	4.15
Optimal Annulus	17.02
Transit Duration (day)	0.0403 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.125 ± 0.023 vs 0.132 (deviation 0.29σ)
Duration fit vs published	0.0403 d vs 0.0954 d (deviation 17.38σ)
Mid-time O–C	-3245.2 min
Depth SNR	5.1
Residual scatter	0.75 %

Light curve

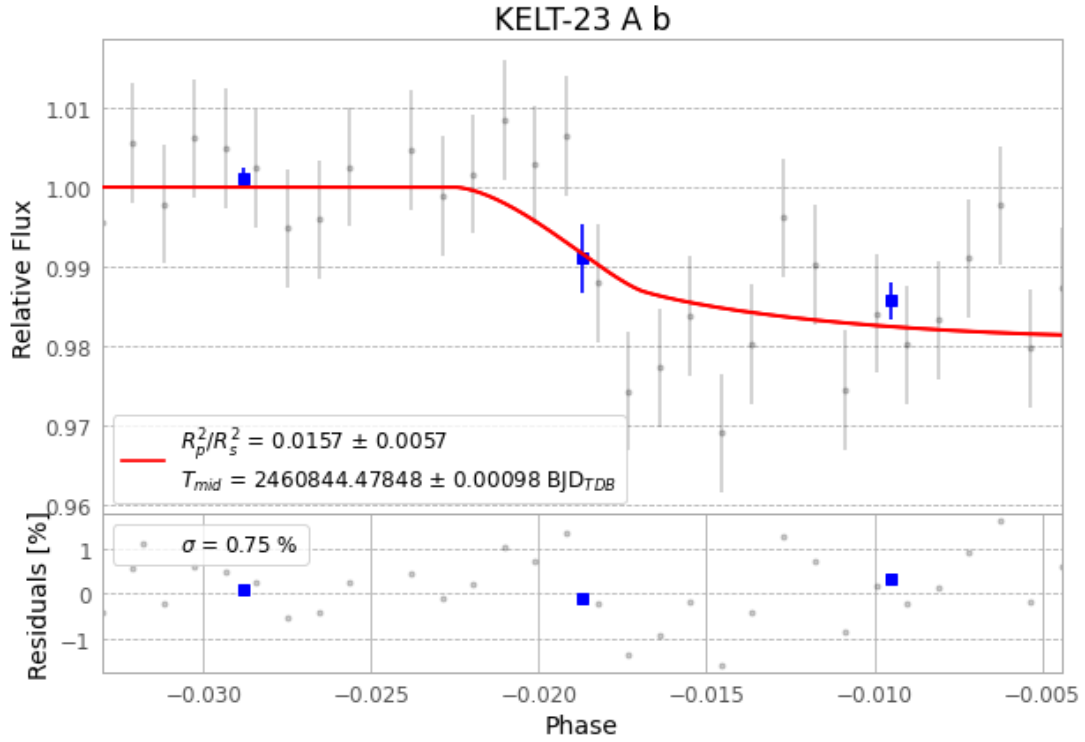


Figure 1 — FinalLightCurve_KELT-23 A b_2025-06-19.png (SHA-256 17c1f9f1e72b4d83...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8ce366f272328f55...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

32 FITS frames (21 MB), KELT-23A250620034817.FITS → KELT-23A250620052116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-23A-250620.log (5a62d7cedb23...), FinalLightCurve_KELT-23 A b_2025-06-19.png (17c1f9f1e72b...), FinalLightCurve_KELT-23 A b_2025-06-19.csv (37c3d48d0693...)

Compute resources

Wall time 86.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 23:56Z.